

PHY201 : Waves and Optics
Second Mid-Semester Examination 2025
Duration: 1 hour

Instructor: A. Chaudhuri

Max. Marks 20

- All questions are compulsory.
- Show intermediate steps clearly for full credit.
- Symbols have their usual meanings.

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1. **[6 marks]** Two identical pendulums of length L are coupled by a spring of coupling strength k . One pendulum is displaced by a small amount and released.
- (a) Derive expressions for the normal mode frequencies. [3]
 - (b) Explain qualitatively how energy is transferred back and forth between the pendulums. [1]
 - (c) If $L = 0.5$ m, $g \approx 10$ m/s², and the coupling shifts one frequency upward by 2%, calculate the beat period. ($\sqrt{20} = 4.472$) [2]

2. **[6 marks]** A stretched string of length 2.0 m and linear density $\mu = 10^{-2}$ kg/m is fixed at both ends and under tension $T = 100$ N. The transverse waves satisfy

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad v = \sqrt{\frac{T}{\mu}}.$$

- (a) Derive the general formula for the frequencies of its normal modes. [3]
 - (b) Compute the frequencies of the first three modes. [2]
 - (c) A tuning fork at 78 Hz excites the string. Which mode will resonate most strongly and why? [1]
3. **[4 marks]** A sinusoidal wave $\xi(x, t) = \xi_0 \cos(kx - \omega t)$ travels down a steel rod of density ρ and Young's modulus Y . For a linear, lossless rod, the longitudinal wave equation is: $\frac{\partial^2 \xi}{\partial t^2} = c^2 \frac{\partial^2 \xi}{\partial x^2}$ where $c = \sqrt{Y/\rho}$. Stress and strain are related by Hooke's law: $\sigma(x, t) = Y\varepsilon(x, t) = Y \frac{\partial \xi}{\partial x}$.
- (a) Derive expressions for the time averaged kinetic (u_K) and potential energy densities ($u_P = \frac{1}{2}\sigma\varepsilon$) of the wave. [2]
 - (b) What is the average energy flux S proportional to? [2]
4. **[4 marks]** Two waves on a string are given by

$$y_1 = 0.50 \cos(20x - 100t), \quad y_2 = 0.50 \cos(20x - 104t),$$

where y and x are in cm and t in s.

- (a) Write the expression for the resultant wave. [2]
- (b) Find the beat frequency and explain its physical significance. [1]
- (c) Determine the maximum transverse velocity of a particle on the string. [1]